

2. Derivable Judgement

Q.1) What's inferential statistics?

Ans Inferential statistics is the process of using data from a sample to draw conclusions or make estimates about a larger population.

Ex: instead of studying every student in a college, we can study a sample of students and use the results to estimate the performance of the entire student populations

Q.2) What is hypothesis testing and its components?

Ans Hypothesis testing is a statistical method used to determine whether sample data provides enough evidence to support or reject a claim about a population. It compares two competing statements using sample data.

- Null Hypothesis (H_0) - assumes there is no significant difference or effect.
- Alternative Hypothesis (H_a) - represents the opposite claim or the effect being investigated.
- Significance level (α) - the maximum acceptable probability of rejecting a true H_0 .
- Test statistics - a value calculated from the sample data to measure evidence against H_0 .
- P-value - indicates how compatible the observed data is with H_0 .
- Decision: Compare the p-value with α : if $p < \alpha$, reject H_0 ; otherwise fail to reject H_0 .

Q-3) Confidence Interval

Ans A confidence Interval is a range of values calculated from sample data that is used to estimate an unknown population parameter. It gives us a range rather than a single estimate along with a chosen confidence level such as 90%, 95%, 99%.

- A critical value is a cut-off value from a probability distribution that determines how far we extend from our sample estimate to create the confidence interval. It depends on the chosen confidence level and the distribution being used.

Q-4) Define P value

- A p value is the probability of obtaining a result as extreme as or more extreme than, the observed result, assuming that the null hypothesis is true.

Q-5) Differentiate Type I and Type II errors.

	Type I	Type II
meaning	Rejecting a true null hypothesis	Failing to reject a false null hypothesis
Also called	False positive	False negative
probability	α	β
Simple idea	Detecting an effect when there is actually no effect	Missing an effect that actually exists.

Q-6) Brief description of Z-test, t-test, chi square test and ANOVA test

Ans/

- Z-test: A hypothesis test based on the standard normal distribution. It is used to test a population mean when the population standard deviation is known, or for proportions under appropriate conditions.
- T-test: A hypothesis test based on the t-distribution, used to compare means when the population standard deviation is unknown.
- Chi-square test: A test based on the χ^2 distribution, commonly used for categorical data, such as testing independence between variables or whether observed frequencies fit an expected distribution.
- ANOVA test: Analysis of variance is used to determine whether the means of three or more groups are statistically equal or whether at least one group mean differs, it uses an F-statistic.

Q-7) What's covariance?

Ans/ Covariance measures how two variables change together. It looks at whether values of X and Y tend to be above or below their respective means at the same time.

Q-8) What's correlation?

Ans/ Correlation measures the strength and direction of the linear relationship between two variables. It's usually represented by the correlation coefficient r , which ranges from -1 to +1.